Electricity Consumption Analysis and Applications based on Smart Grid Big Data

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***Abstract*— With the background of active economies and rapid developments in Shanghai, it is desirable to give an insight into the huge electrical consumption data that is already collected from customers and stored in the electrical vendor side. These data is accumulating explosively due to the Smart Grid deployment of pervasive intelligent instruments. If the implicit information of the data could be figured out, for example, the customer behaviors in relation with energy consumptions, it will benefit the electrical company to optimize the energy offering efficiency and upgrade the customer service level. This paper presents parts of the results of the "Big Data Technical Research and Application on Customer Side" project conducted in State Grid Shanghai Municipal Electric Power Company (SMEPC). The results include detection and correction method of data outliers, characteristic analysis of customer behavior, correlation between different industries, and potential commercial application of smart grid data. It is believed that the application of big data in smart grid is quite appealing.**

Keywords - big data; smart grid; electricity consumption

I. INTRODUCTION

With the development of smart grid and the high degree integration of power industry and information technology, large amounts of data ranging from power generation, transmission, consumption, and real-time status monitoring, to the climate and geography data have been produced and collected in storage. These data is accumulating explosively due to vastly deployed sensors such as phase measurement units (PMUs) and smart meters etc. These data is desired to be collected, aggregated, and delivered to analytical applications to support control decision of promoting the efficiency and reliability of power grid. An market analysis report " The Soft Grid 2013-2020:Big Data & Utility Analytics for Smart Grid" has presented a taxonomy that identifies three major domains in which analytics can aid utilities, which are enterprise analysis, grid operation analysis and customer analysis [1]. All are ripe with opportunity. Related academic researches include transient power quality assessment [2], devices failure detection [3], prediction of energy consumption [4][5] etc. Contrast to academic thriving, most of the electrical utilities claimed underprepared to manage smart grid data influx [6], expect electrical company ComEd in California has published their findings in exploring opportunities of big data analysis [3].

As previously described, customer analysis is one of the most appealing domains for utilities to drive great value from smart grid data. The energy consumption data on customer side implies the customer behavior characteristic, which could be useful for electricity consumption feature analysis of consumers in different districts and with various industries, classification of consumers and individualized power services, electricity price decision support, and other value-added services[7][8][9][10]. Generally, the majority of present researches are at the feasibility study stage. Thus, how to figure out the implicit information in energy consumption data by big data technique and thoughts, and to support electrical companies in optimizing the energy offering efficiency and improving the customer service level, is one of the most notable research spots for power utilities。

This paper presents parts of the results of the "Big Data Technical Research and Application on Customer Side" project conducted in State Grid Shanghai Municipal Electric Power Company (SMEPC), based on power consuming data of nearly one million customers including residential, industrial and commercial consumers. In the following text, the research framework is firstly introduced in section 2. Then the method of data preprocessing is demonstrated in section 3, and the data analysis and applications in section 4. The transferability and influence factors of this study are discussed in section 5. And finally, the paper is concluded with the profound contribution of big data idea in energy industry.

II. ELECTRICITY CONSUMPTION ANALYSIS AND APPLICATION RESEARCH FRAMEWORK

Figure 1 shows the research framework of this paper, which consists of three parts, i.e. data collection, data storage and preprocessing, data mining and applications. Data sources in this paper include power consumption data, electricity consumption behavior of customers, social and economic data, geographical data etc. Data collection could be carried out by the automatic data collection devices, such as smart meters and power load control systems, or by other non-automatic ways, such as social surveys and public data bases. Those data are transferred to distributed storage system of Hadoop clusters and then indexed and retrieved by Hbase data base. Based on these data, outlier detection and data interpolation are implemented in the data pre-processing step. In order to uncover the implicit values of the data, data mining techniques are adopted. The correlation analysis methods and prediction algorithms are used to support the application analysis.

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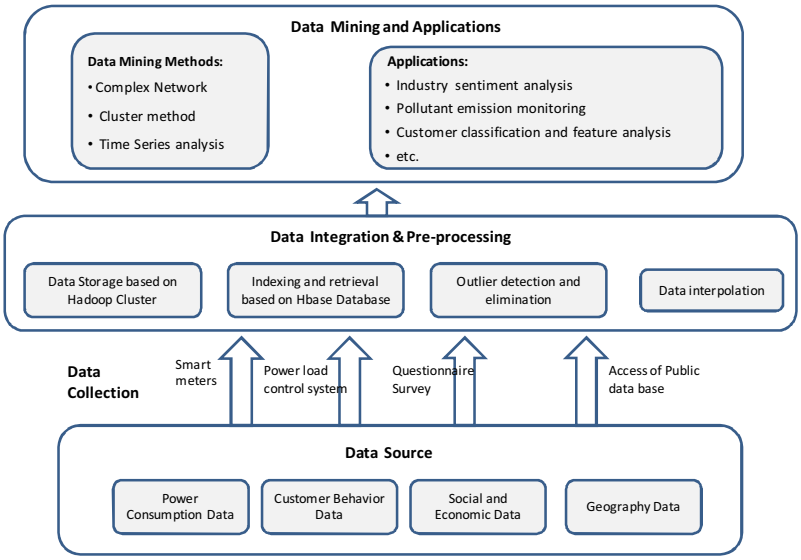


Figure 1. Electricity consumption analysis and application research framework based on big data

III. DATA PRE-PROCESSING

The power consumption data in the research are sampled from 800,000 customers located in different districts of Shanghai city, and comprised by all the industrial and commercial enterprises and 15% residents of the whole city.

The power data utilized in this research are unevenly distributed, being lack of information and mixed with outliers. Therefore, the data preprocessing has significant impact on the research results. The main concepts adopted in preprocessing include detecting those data larger than 100 times of the average as outliers, and modifying them by global linear interpolation methods.

The flow charts are shown in Figure 2 and Figure 3. The benefit of this preprocessing is that the main industry characteristics of power consumption have remained significantly.

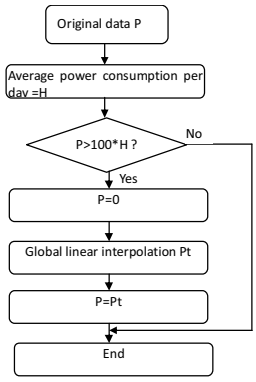


Figure 2. Pre-processing

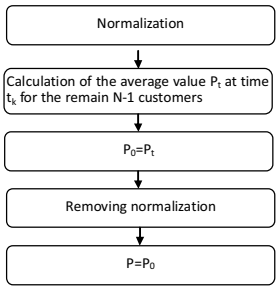


Figure 3. Gobal linear interpolation

In terms of the interpolation processing, the traditional approach is to calculate the average of customer power consumption and then make the interpolation with this value. While in this paper, the whole industry power consumption features are taken into consideration. The concept of global linear interpolation is to retain the most industry implicit information. First, the normalization of the whole industry values is carried out. Then, in order to interpolate the power consumption value P_t at time t for customer N, the average value of other N-1 customers $\overline{P_t}$ at time t is calculated, where N is the total number of customers in the industry. Second, the normalization is removed and the final result is interpolated to the missing value or outliers. Compared with the traditional local linear interpolation, the global liner interpolation method takes advantage of the whole industry electricity consuming features, by combination of the individual consuming statistic scale.

IV. APPLICATION CASES OF BIG DATA ANALYSIS

A. *Industry Prosperity analysis*

Presently, the industry customers make the majority of the whole electricity consuming market. Their power consumption condition exerts a significant influence to the total social electricity usage. Generally, the power consumption amount is proportional to the productivity of industrial enterprises. Therefore, this paper investigates the industry sentiment condition from the power consumption data view, and further understands the status quo of city economic structure, so as to support the government decision.

- "Booming index" of Industry

Intuitively, if the power consumption amount of an industrial customer is in a rising trend, which is considered as a snapshot of its sentiment condition, the fitting line of its electricity curve is upwardly sloping. Thus, the slope coefficient can be considered as a sentiment index, so called "booming index".

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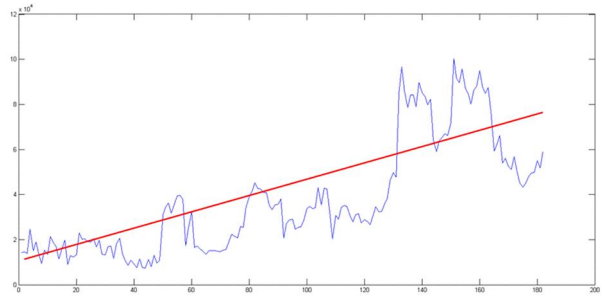


Figure 4. Power consumption curve (blue) with fitting line (red). X axis denotes weeks from 1/1/2007 to 27/6/2010, and Y axis power consumption amount (kwh)

Figure 4 shows the power consumption curve of an industrial customer and its best fitting line. The slope of the fitting line is larger than 0, indicating the thriving development of this enterprise.

Figure 5 shows the "booming index" of 293 industrial customers in Shanghai city. It demonstrates that the insurance, the railway industry and the technical service are in thriving status, while the wood products and social cultural industry are developing relatively slow.

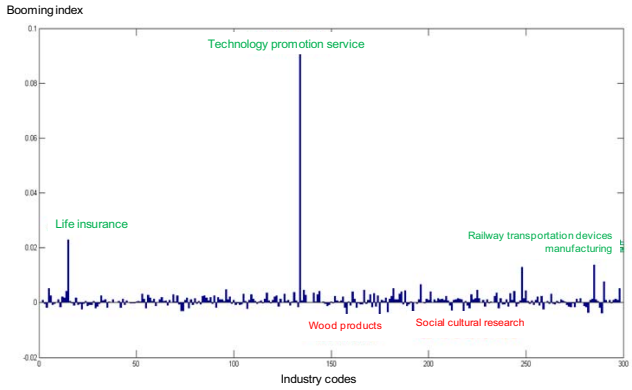


Figure 5. "booming index" of 293 industries in Shanghai city

- Industry community analysis

In order to explore the associations of electrical consumption behaviors between different industries, the clustering algorithms and correlation analysis are adopted for a statistical analysis of similarity. Those industries with similar electrical consumption behaviors are grouped together to form a community. Different from the traditional classification standard based on the artificially defined national industry class, this paper is just relying on the electrical consumption data to explore the community structure, which contains the implicit correlations between different Shanghai industries. The correlations could be of a cluster (or decentralization) or promotion (or constraints), which may provide a reference for the industry classification and city economic structure analysis.

In this paper, all industrial and business customers with various power voltages were taken into research. Those industrial customers with sampling points less than 100 per year are deleted, and then correlation analysis is conducted for the rest, according to their power consumption data to get correlation matrix. Finally, according to the correlation coefficient of the resulting matrix, threshold is set for classification of communities. Results showed that only small industrial and business clients with power supply voltage of 220V have the distinct community attribute.

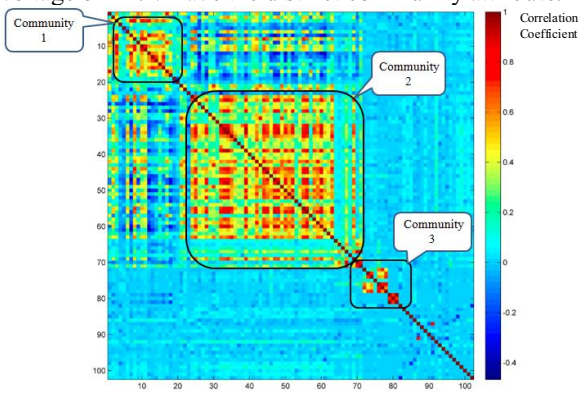


Figure 6. The Industry Communities based on Similar Electrical Consumption（The x axis and y axis are industry codes. Color represents the value of the correlation coefficient, the colder colors represents lower correlation.）

Figure 6 shows communities formed by small industrial and business clients according to the similar power consumption features. The x axis and y axis are industry codes. Color represents the value of the correlation coefficient, a warmer colors represents a higher correlation. Obviously, there are three major communities. Community 1 consists of customers from traditional primary industry and secondary industry, e.g. manufacturing industry, agriculture and construction fields. Community 2 covers customers from service industries, including gardening, transportation industry, technology promotion service, lighting, warehousing, postal service and internet information etc. Community 3 is constituted by enterprises and organizations from education industry, culture, public health and welfare.

The industry community analysis results demonstrate that the economic structure of Shanghai city is mainly dependent on the service industries and supplemented by the manufacturing industry and agriculture. This result is consistent with the present situation. Therefore, the industry sentiment analysis and urban economic development evaluation from power data view using big data techniques, is proved to be promising.

B. Pollutant emission monitoring

Generally, the power consumption amount is proportional to the productivity of an industrial enterprise, while the productivity is also positive correlated to its pollutant emission. Thus, the pollution detection could be performed by monitoring the electricity consuming, and further a pollutant emission monitoring mechanism could be established.

In this study, the quantitative relationship between electricity use, industrial production and pollutant emissions is analyzed for large and medium size industrial enterprises.

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On this basis, a pollutant emissions monitoring mechanism could be established to predict the emission based on the power usage data collected from smart meters in real time. If the pollutants are emitted excessively, some administrative measures are suggested to take in order to keep the overall air condition at stable level. The measures could include the power supply control, load shifting and other means of regulating and encouraging industrial companies to change production plans.

The pollutant emission monitoring in this paper is conducted in three steps.

Step 1: Get primary energy consumption data of industrial enterprises, then calculate the emissions of air pollutants based on these data by using the coal-fired emission factor, fuel emission factors of each specific industry.

Step 2: Based on the pollutant emission data and electrical consumption data during a period, the pollutant coefficients of pollutant emission per electricity usage is calculated.

Step 3: With the pollutant coefficients and electrical consumption monitoring data (e.g. sampled at 15 minutes period), the air pollution curves can be depicted to show the real time pollution level at certain districts.

C. *Analysis of Resident's Electrical Consumption*

The analysis includes two aspects. The first one studies the amount of electrical consumption. In this kind of study, the residents within a district are divided into three categories, i.e. normal consumption, small amounts of consumption and big amounts of consumptions. On another aspect, the electrical consumption behaviors are studied to find out user preferences. In this kind study, the residents are categorized into two types, normal consumption and abnormal consumption. Then based on the normal consumption data, the residents sensitivity to the climate temperature can be figured out into three levels, i.e. insensitive, neutral and sensitive to the temperature changes

- "quantitative index" of electrical consumption

The "quantitative index" is calculated by: first, calculate each family's average daily electrical consumption within a district of city during a period of 538 days. Then the median value can be calculated over all the residents in this district. This median value is called "Quantitative Index" of electrical consumption, denoted as IndexQ. With this "Quantitative Index", all the residents in this district could be categorized into three types. Those families whose average daily electrical consumption value larger than 2 times of IndexQ is considered as high consumption families, while those smaller than 0.5 times of IndexQ are low consumption families. The rest are the normal.

TABLE I. "QUANTITATIVE INDEX" AND USER CATEGORIES

User Types / Zones	Quantitative Index	Family Counts of Low consumption	Family Counts of High consumption	Family Counts of Normal consumption
Zone 0	5.74	65003	50897	104664
Zone 1	5.74	76914	61585	115199
Zone 2	6.06	61134	49554	91127
Zone 3	5.53	33642	29198	40288

Notes: **Zone 0**: city center zone; **Zone 1**: Btw city center and outer circle; **Zone 2**: Btw outer circle and suburban area; **Zone 3**: Outside suburban area

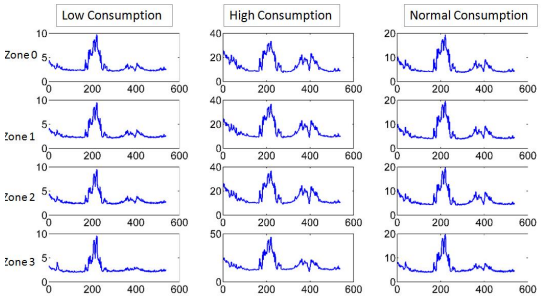


Figure 7. Power Consumption Curves of All Districts Users Categorized by different Quantitative Index (the x-axis is the date period of totally 538 days, Y-axis is electricity consumption amount in kWh)

In Table 1, the IndexQ values of different districts do not have big gaps. That means the residents in urban and suburban districts have the similar electrical consumption behaviors. In Figure 7, it is prominent that the electricity consuming amount reaches the peak level at "200"and "400"of the x-axis. These two points are actually in the summer and winter.

- "Sensitive Index" of Temperature

First, the average daily electrical consumption of each family within a district of city is calculated for a period of 2 years. Second, the electrical consumption of each family is divided by the average, and the result is called the unit average electrical consumption (Uaec), which is useful to show the feature of power consuming behavior. Then, calculate Uaec in three periods respectively that are winter, summer, spring and autumn, denoted by Uaec_winter, Uaec_sumer, and Uaec_spr_aum. Next, Uaec_winter and Uaec_sumer are divided by Uaec_spr_aum respectively, and the results are named by winter sensitive index (Index_sensitive_winter) and summer sensitivity index (Index_sensitive_summer). The product of these two indices is sensitive index of temperature, namely Index_sensitive_ws. The median value of Index_sensitive_ws indicates the sensitive extent of residents in a district to climate temperature, namely IndexS. With this IndexS, all the residents could be categorized into three types. Those families whose Index_sensitive_ws value larger than 2 times of IndexS is considered as sensitive families, while those smaller than 0.5 times of IndexS are insensitive families. The rest are the neutral.

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TABLE II. "SENSITIVE INDEX" OF TEMPERATURE AND USER CATEGORIES

User Types / Zones	Sensitive Index	Family Counts of Insensitive users	Family Counts of Sensitive users	Family Counts of Neutral users
Zone 0	3.06	6811	22180	180837
Zone 1	2.95	12255	29375	201868
Zone 2	2.66	12696	25987	156174
Zone 3	2.56	5558	10842	81352

Notes: **Zone 0**: city center zone; **Zone 1**: Btw city center and outer circle; **Zone 2**: Btw outer circle and suburban area; **Zone 3**: Outside suburban area

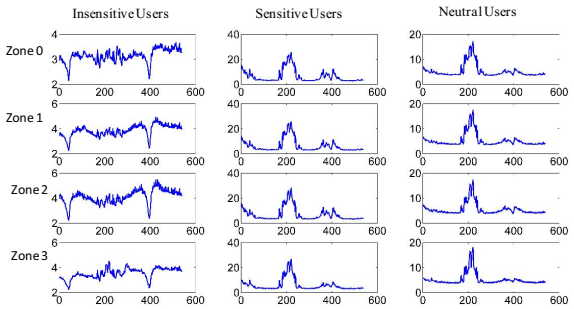


Figure 8. Electrical Consumption Curves of All Districts Users Categorized by different Sensitive Index (the x-axis is the date period of totally 538 days, Y-axis is electricity consumption amount in kWh)

In Table 2, the "Sensitive Index" of Temperatures takes a decreasing trend. This means that the closer to the center of Shanghai city, the higher electricity consumption in winter and summer, and higher frequency of using air conditioning and warming appliances. In Figure 8, there are some users (insensitive customers) whose average electricity consumption in summer and winter is no more than in spring and autumn or even less. For most users (ordinary users), the average electricity consumption in summer are significantly higher than in spring and autumn seasons but not obvious in winter. There are still some users (sensitive ones), their electricity consumption in both summer and winter are significantly higher than in other seasons. These users generally tend to use large-scale cooling (heating) appliances in uncomfortable weather, and may be potential energy-saving clients. In future studies, we will make further characterization research of the user's personality traits including family structure, education level, types and energy consumption of household appliances, by social surveys and behavior analysis methods, so as to explore the smart management measures on electricity demand-side.

V. DISCUSSION

A. *Transferability*

In this paper, the electricity consumption analysis and application research framework based on big data is built, which consists of three parts, i.e. data collection, data storage and preprocessing, data mining and applications. Different from traditional analysis framework, we use the distributed storage system of Hadoop clusters, and Hbase-based indexing and extraction methods, improving the efficiency of data storage and indexing. The whole framework is complete and versatile, adaptable to other applications of big data analysis.

B. *Influence factors*

There are two significant influence factors of big data analysis in this research. One is "Big Data thought", the other is the data quality.

It is said that the essence of big data technologies, focuses on correlation analysis instead of the traditional causation. Based on the positive correlation between electricity production and industry behavior and pollution behavior, this paper confirms the characteristics of economic behaviors from the power data view. For example, the "booming index" reflects the industry booming degree. Therefore, carrying out relevant analysis, and cultivating big data thought, is quite important in big data analysis.

Another key factor is the quality of the data itself. Since the electric power company puts more attention to the monthly accumulated data, the acquisition and transmission of daily data from the smart meters tends to be less valued, and transmission errors and omissions occur occasionally. Although the global data interpolation method can be helpful to fill the missing data, expert experiences and historical data is still needed for validation, which costs nearly half of all research over time. Therefore, improving the data quality and managing the data as an important corporate assets, establishes important foundation for the traditional industries such as power companies to carry out big data strategy.

VI. CONCLUSION

This paper leverages the big data analysis platform and data mining methods to study the electrical consumption features and associations of industrial and commercial customers and the residents of Shanghai city. The industry sentiment analysis and energy consumption analysis of residents are conducted and provide the testimony of Shanghai economical features. The research methods could be used to improve the grid management on demand side and guide users' energy consumption behaviors by leveraging the big data techniques.

As one of the fundamentals of the Third Time Industrial Revolution, the smart grid would play an important role and promote the global energy evolution. An "Internet of Energy" is desired to come up with Strong, Interconnected, Intelligent and Interactive features. Some interesting challenges exist not only in the management of the future smart grid data, but also on how to extract grate value from this data. Future work could explore more novel business models of smart grid based on the availability of big data.

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